ESS (EMERGENCY STOP SIGNAL SYSTEM) OPERATES WHILE NOT UNDER OPERATION CONDITIONS [EMERGENCY STOP SIGNAL SYSTEM (ESS)]

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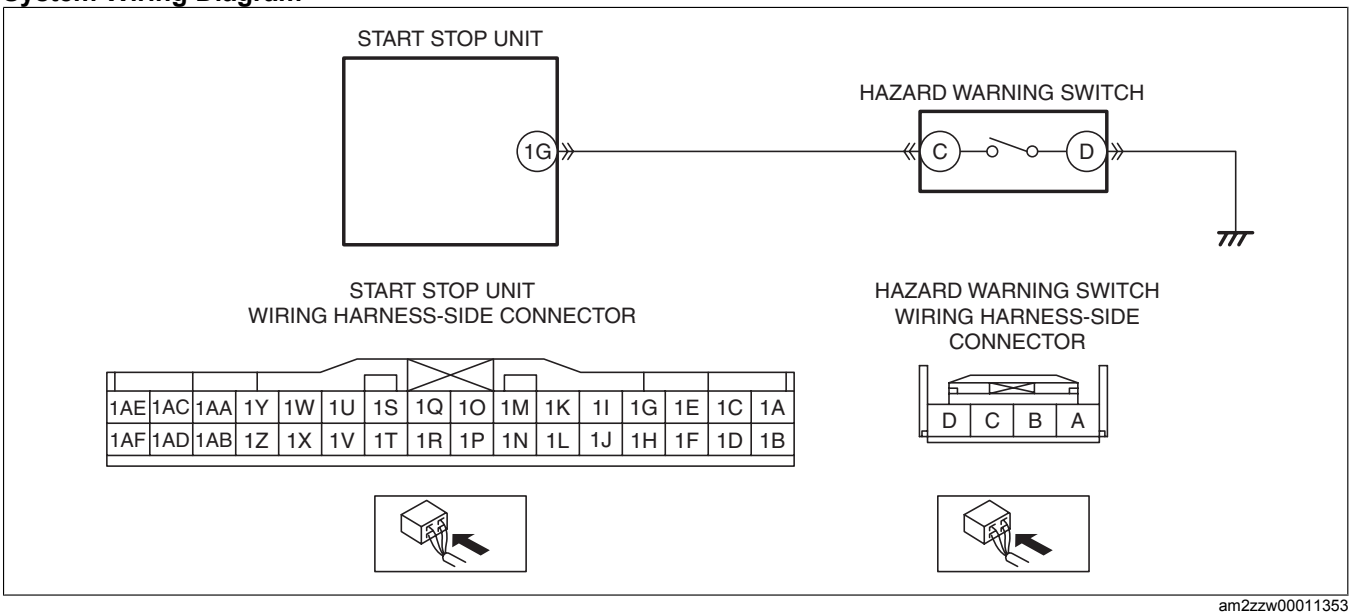
Note
- If there is any vehicle malfunction complaint lodged by a customer, perform FOREWORD [EMERGENCY STOP SIGNAL SYSTEM (ESS)] malfunction diagnosis according to the troubleshooting procedure. (See FOREWORD [EMERGENCY STOP SIGNAL SYSTEM (ESS)].)

Description
- Hazard warning lights flash while driving.
- Hazard warning lights flash even though deceleration is done without decelerating rapidly.

Possible Cause
Hazard warning light flash malfunction
- Hazard warning switch malfunction (temporary short circuit caused by shock from road surface)
  - — Hazard warning switch malfunction
  - — Short to ground in wiring harness between hazard warning switch terminal C and start stop unit terminal 1G

Sudden deceleration wheel speed is detected
- Grip level of tires is low
  - — Driving on a low-traction surface
  - — Tire wear (hydroplaning)
- Start stop unit malfunction
- BCM malfunction

System Wiring Diagram



am2zzw00011353

Diagnostic Procedure
- *When performing an asterisked(*) troubleshooting inspection, first verify the damage and connection status of the connectors and terminals, then perform the inspection while shaking the wiring harness to discover whether poor contact points are the cause of an intermittent malfunction. If there is a malfunction in the connectors or terminals, securely connect, repair, or perform replacement.

Step	Inspection		Action
1*	**DETERMINE IF MALFUNCTION CAUSED BY HAZARD WARNING SWITCH SIGNAL ERROR** • Display the start stop unit PID HAZARD_SW item using the M-MDS. (See PID/DATA MONITOR INSPECTION [START STOP UNIT].) • Verify the PID/data monitor display while operating the hazard warning switch. • Does the PID/data monitor display switch?	Yes	Go to Step 4.
		No	Go to the next step.

Step	Inspection		Action
2	**INSPECT HAZARD WARNING SWITCH FOR MALFUNCTION** • Remove the hazard warning switch. (See HAZARD WARNING SWITCH REMOVAL/ INSTALLATION.) • Inspect the hazard warning switch. (See HAZARD WARNING SWITCH INSPECTION.) • Is the hazard warning switch normal?	Yes	Go to the next step.
		No	Replace the hazard warning switch. (See HAZARD WARNING SWITCH REMOVAL/ INSTALLATION.)
3*	**INSPECT FOR SHORT TO GROUND IN WIRING HARNESS BETWEEN HAZARD WARNING SWITCH AND START STOP UNIT** • Inspect for continuity between hazard warning switch terminal C and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between hazard warning switch terminal C and body ground. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit.
		No	Replace the start stop unit. (See START STOP UNIT REMOVAL/INSTALLATION.)
4*	**INSPECT BCM FOR MALFUNCTION** • Display the BCM PID HAZARD_SW item using the M-MDS. (See PID/DATA MONITOR INSPECTION [BCM].) • Verify the PID/data monitor while operating the hazard warning switch. • Does the PID/data monitor display switch?	Yes	Go to the next step.
		No	Replace the BCM. (See BODY CONTROL MODULE (BCM) REMOVAL/ INSTALLATION.)
5	**INSPECT TIRES FOR MALFUNCTION** • Inspect the tires for the following: — Tire air pressure (are all four tires at recommended pressure?) — Tire wear condition (is tire wear uniform on all four tires, or is it different?) — Tire size • Are the tires normal?	Yes	Phenomenon occurring due to effect of road surface can be considered. (Emergency Signal System (ESS) normal) If the malfunction recurs, replace the BCM. (See BODY CONTROL MODULE (BCM) REMOVAL/ INSTALLATION.)
		No	Adjust the malfunctioning tire or replace the tire.